Homework 4

Due Thursday, September 24, 2026 by 11:59 PM on Gradescope

Instructions

- Include your full name and Coyote ID on the top right of the first page
- Clearly circle or box your answer for any question involving calculations. If the problem is multipart, circle or box answers for each part.
- Work should be neatly written and easy to follow.
- You must tag each question on Gradescope. If you do not tag your work for a given problem, you will not receive credit for it.

Problems

This homework contains problems related to 2D kinematics, including projectiles, uniform circular motion, and nonuniform circular motion.

1. A ball thrown horizontally at $25\,\mathrm{m/s}$ travels a horizontal distance of $50\,\mathrm{m}$ before hitting the ground. From what height was the ball thrown?
2. In the Olympic shootout event, an athlete throws the shot with an initial speed of $12.0\,\mathrm{m/s}$ at a $40.0°$ angle from the horizontal. The shot leaves her hand at a height of $1.8\,\mathrm{m}$ above the ground. How far does the shot travel?
3. On the Apollo 14 mission to the moon, astronaut Alan Shepard hit a golf ball with a 6 iron. The free-fall acceleration on the moon is $1/6$ of its value on earth. Suppose he hit the ball with a speed of $25\,\mathrm{m/s}$ at an angle $30°$ above the horizontal
   a. How much farther did the ball travel on the moon than it would have on earth?
   b. For how much more time was the ball in flight?
4. A boat takes $3.0\,\mathrm{hours}$ to travel $30\,\mathrm{km}$ down a river, then $5.0\,\mathrm{hours}$ to return. How fast is the river flowing?
5. Susan, driving north at $60\,\mathrm{mi/h}$, and Trent, driving east at $45\,\mathrm{mi/h}$, are approaching an intersection. What is Trent's speed relative to Susan's reference frame?
6. An old-fashioned single-play vinyl record rotates on a turntable at $45\,\mathrm{rev/min}$. What are
   a. The angular velocity in $\mathrm{rad/s}$?
   b. The period of the motion?

7. A $3000\,\mathrm{m}$ high mountain is located on the equator. How much faster does a climber on top of the mountain move than a person at the base of the mountain (assume they are at sea level)? The earth's radius is $6400\,\mathrm{km}$.
8. To withstand "g-forces" of up to $10\,g$'s, caused by suddenly pulling out of a steep dive, lighter jet pilots train on a "human centrifuge." $10\,g$ is an acceleration of $98\,\mathrm{m/s^2}$. If the length of the centrifuge arm is $12\,\mathrm{m}$, at what speed is the rider moving when she experiences $10\,g$'s?
9. The radius of the earth's (very nearly) circular orbit around the sun is $1.5 \times 10^{11}\,\mathrm{m}$. Find the following magnitudes of the earth as it travels around the sun (assume a year is $365\,\mathrm{days}$)
   a. Velocity
   b. Angular velocity
   c. Centripetal acceleration
10. A $5.0\,\mathrm{m}$ diameter merry-go-round is initially turning with a $4.0\,\mathrm{s}$ period. It slows down an stops in $20\,\mathrm{s}$
    a. Before slowing, what is the speed of a child on the rim?
    b. How many revolutions does the merry-go-round make as it stops?
11. An electric fan goes from rest to $1800\,\mathrm{rev/min}$ in $4.0\,\mathrm{s}$. What is its angular acceleration?

12. A cannonball is fired at $100\,\mathrm{m/s}$ from a barrel tilted upward at $25°$.
    a. Find an expression for the cannonball's direction of travel, measured as an angle from horizontal, after traveling a distance d horizontally.
    b. How far has the cannonball traveled horizontally when it reached its maximum height?
    c. What is the angle after the cannonball travels $500\,\mathrm{m}$?
13. A typical laboratory centrifuge rotates at $4000\,\mathrm{rev/min}$. Test tubes have to be placed into a centrifuge very carefully because of the very large accelerations.
    a. What is the acceleration at the end of a test tube that is $10\,\mathrm{cm}$ from the axis of rotation?
    b. For comparison, what is the magnitude of the acceleration a test tube would experience if dropped from a height of $1.0\,\mathrm{m}$ and stopped in a $1.0\,\mathrm{ms}$ long encounter with a hard floor?
14. Astronauts use a centrifuge to simulate the acceleration of a rocket launch. The centrifuge takes $30\,\mathrm{s}$ to speed up from rest to its top speed of 1 rotation every $1.3\,\mathrm{s}$. The astronaut is strapped into a seat $6.0\,\mathrm{m}$ from the axis.
    a. What is the astronaut's tangential acceleration during the first $30\,\mathrm{s}$?
    b. How many g's of acceleration does the astronaut experience when the device is rotating at top speed? Each $9.8\,\mathrm{m/s^2}$ of acceleration is $1\,g$.

15. A computer hard disk $8.0\,\text{cm}$ in diameter is initially at rest. A small dot is painted on the edge of the disk. The disk accelerates at $600\,\text{rad/s}^2$ for $1/2\,\text{s}$, then coasts at a steady angular velocity for another $1/2\,\text{s}$.
    a. What is the speed of the dot at $t = 1.0\,\text{s}$?
    b. Through how many revolutions has the disk turned?
16. Flywheels—rapidly rotating disks—are widely used in industry for storing energy. They are spun up slowly when extra energy is available, then decelerate quickly when needed to supply a boost of energy. A $20\,\text{cm}$ diameter rotor made of advanced materials can spin at $100{,}000\,\text{rev/min}$.
    a. What is the speed of a point on the rim of this rotor?
    b. Suppose the rotor's angular velocity decreases by $40\,\%$ over $30\,\text{s}$ as it supplies energy. What is the magnitude of the rotor's angular acceleration? Assuming that the angular acceleration is constant.
    c. How many revolutions does the rotor make during these $30\,\text{s}$?
17. In your opinion, what was the hardest question of this homework. What was the easiest? What made the problem hard or easy?